TWO CHOICES OF SPLITTING GIVE HOMOTOPIC CONNECTING MORPHISMS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file homology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument; both assertions are written out, each with its displayed computation, and the closing sentence is a conclusion rather than a placeholder.

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Editorial note. The same-name check for this card scanned every staged Stacks chapter of this repository – all sources/stacks-*/pinned-original-source.tex files, the card's own chapter excluded – and the labels found are recorded in dedup.same_name_cross_chapter on the card.

Editorial note. The preamble is the chapter's own native preamble excerpt already used for this source: the comment-type reference, slogan and history environments (native preamble lines 8-13), the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217, the actual last line of the file).

Allowed standard prerequisites (editorial).

- The connecting morphism of a termwise split sequence. Same chapter, Lemma lemma-ses-termwise-split-cochain, delivered on this card as a context span. Supplies the notation, the hypotheses and the constructed morphism delta whose dependence on the choice of splitting is the subject of the lemma.
- Homotopy of morphisms of complexes. Same chapter, the definition of a homotopy between two morphisms of complexes. Supplies the meaning of the second assertion, namely the displayed formula expressing delta' minus delta as a boundary.

Lemma 0.1. *Let $\mathcal{A}$ be an additive category. Let*

$$0 \to A^\bullet \to B^\bullet \to C^\bullet \to 0$$

be a complex (!) of complexes. Suppose that we are given splittings $B^n = A^n \oplus C^n$ compatible with the maps in the displayed sequence. Let $s^n : C^n \to B^n$ and $\pi^n : B^n \to A^n$ be the corresponding maps. Then the family of morphisms

$$\pi^{n+1} \circ d_B^n \circ s^n : C^n \to A^{n+1}$$

define a morphism of complexes $\delta : C^\bullet \to A[1]^\bullet$.

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Lemma 0.2. *Notation and assumptions as in Lemma 0.1. Let $\alpha : A^\bullet \to B^\bullet$, $\beta : B^\bullet \to C^\bullet$ be the given morphisms of complexes. Suppose $(s')^n : C^n \to B^n$ and $(\pi')^n : B^n \to A^n$ is a second choice of splittings. Write $(s')^n = s^n + \alpha^n \circ h^n$ and $(\pi')^n = \pi^n + g^n \circ \beta^n$ for some unique morphisms $h^n : C^n \to A^n$ and $g^n : C^n \to A^n$. Then*

(1) *$g^n = -h^n$, and*
(2) *the family of maps $\{g^n : C^n \to A[1]^{n-1}\}$ is a homotopy between $\delta, \delta' : C^\bullet \to A[1]^\bullet$, more precisely $(\delta')^n = \delta^n + g^{n+1} \circ d_C^n + d_{A[1]}^{n-1} \circ g^n$.*

Proof. As $(s')^n$ and $(\pi')^n$ are splittings we have $(\pi')^n \circ (s')^n = 0$. Hence

$$0 = (\pi^n + g^n \circ \beta^n) \circ (s^n + \alpha^n \circ h^n) = g^n \circ \beta^n \circ s^n + \pi^n \circ \alpha^n \circ h^n = g^n + h^n$$

which proves (1). We compute $(\delta')^n$ as follows

$$(\pi^{n+1} + g^{n+1} \circ \beta^{n+1}) \circ d_B^n \circ (s^n + \alpha^n \circ h^n) = \delta^n + g^{n+1} \circ d_C^n + d_A^n \circ h^n$$

Since $h^n = -g^n$ and since $d_{A[1]}^{n-1} = -d_A^n$ we conclude that (2) holds. □